



Roman Stepanov

Dalby (Lund municipality), Sweden

■ roman.stepanov@beltandbraces.se | ■ +46 76 76 211 21

■ [LinkedIn](#)

Profile

Senior Software Engineer with **25+ years of experience** in **embedded systems, Android, Linux kernel and low-level debugging**.

Strong background in **automotive, telecom, AI acceleration and performance analysis**.

Specialist in **deep root-cause analysis** of complex, safety-critical problems in cross-functional environments.

Available for **B2B consulting or employment**.

Key Competencies

- Embedded Linux & Android (AOSP, BSP, system services)
- Low-level debugging & root-cause analysis
- Automotive platforms (QNX, NVIDIA Drive OS)
- Performance & power profiling
- Kernel, drivers, system bring-up
- AI HW / NPU evaluation

Professional Experience

Sigma Connectivity AB — Lund

Senior Software Engineer (Consultant)

Jan 2025 – Present

- **Consolidating legacy data-collection variants into one unified, multi-platform telemetry system** in AOSP
- **Migrated the codebase from AOSP 14 to AOSP 16** and onboarded existing wearable devices
- Accelerated delivery and code quality with **AI-assisted tools** (Claude, ChatGPT, Copilot)

Tech: AOSP, Clang, LLDB, Soong, Python

Volvo Cars — Lund

Senior Software Developer (Consultant)

Dec 2023 – Jan 2025

- **Delivered performance & debugging tooling** for the car's **High Performance Computing Units (HPCU)** — per-process CPU, memory and core-scheduling visibility
- **Enabled regression/trend tracking across system versions** via comparable captured sessions; surfaced hidden CPU spikes (e.g. on car-door open)

Platforms: ARM / AArch64, NVIDIA Drive OS, QNX

Tech: C/C++, Python, PostgreSQL, Plotly

Sigma Connectivity AB — Lund

Senior Software Developer (Consultant)

2019 – 2023

- R&D for **Meta** on AI-accelerated wearables (EMG hand-gesture sensor, CV/AR); **benchmarked NPU/APU performance & power**, driving platform decisions
- **Delivered the factory test & calibration backend** for a **Microsoft dual-screen Android tablet**, preventing scrapping of finished units on the production line

- Built tracing/profiling tools, display-stack control and system diagnostics

Platforms: ARM, ARM Ethos-U, Qualcomm

Tech: C/C++, Python, TensorFlow, Perfetto, Systrace, OpenCV

Axis Communications — Lund

Senior Software Developer

2018 – 2019

- Linux kernel development for **secure camera platforms**
- **Enabled pre-silicon bootloader debugging** with QEMU peripheral models (I2C, Flash); device drivers, kernel debugging, board bring-up

Tech: C/C++, GDB, QEMU

TerraNet AB — Lund

Senior Software Developer

2017 – 2018

- **Proved a 3M Wi-Fi-mesh headset concept viable** (~10 units simultaneously)
 - Brought a heavyweight Android wearable platform within power budget via CPU & power optimization
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Sony Mobile Communications / Sony Ericsson — Lund

Senior Software Engineer

2008 – 2015

- Member of **MIB (Men In Black)** elite troubleshooting team
- **Saved a large operator order** by root-causing broken video streaming; tamed memory corruption, leaks and watchdogs
- Built internal debugging tools (memory tracker, GDB crash-dump plugin, modem ring-buffer logger)

Platforms: Qualcomm, OMAP, Broadcom, OSE

Education

Riga Aviation University — 1994 – 1998

Faculty of Radioelectronics and Computer Systems

Technical Skills

- **Languages:** C, C++, Python, Java, assembly
 - **OS:** Linux, Android / AOSP, FreeBSD, QNX
 - **Debugging:** GDB, LLDB, Trace32, J-Link, crash/ramdump analysis
 - **Profiling:** Perfetto, Systrace, ftrace, power profiling
 - **Low-level:** Kernel, drivers, board bring-up, DSI/MIPI, I2C, USB, PCI
 - **AI/HW:** ARM Ethos-U NPU's, TensorFlow, PyTorch, OpenCV
 - **Tools:** Git, Gerrit, QEMU, JTAG, Soong, CMake
 - **Architectures:** ARM, AArch64, x86
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Interests

Electronics · Shortwave amateur radio · Software-defined radio (SDR)
